(iii) **When the force and displacement are perpendicular ($\theta = 90^o$) to each other.** when a body moves on a horizontal direction, the gravitational force (mg) does no work on the body, since it acts at right angles to the displacement as shown in Figure 4.3(b). In circular motion the centripetal force does not do work on the object moving on a circle as it is always perpendicular to the displacement as shown in Figure 4.3(c).

For a given force (F) and displacement (dr), the angle (θ) between them decides the value of work done as consolidated in Table 4.1.

There are many examples for the negative work done by a force. In a football game, the goalkeeper catches the ball coming towards him by applying a force such that the force is applied in a direction opposite to that of the motion of the ball till it comes to rest in his hands. During the time of applying the force, he does a negative work on the ball as shown in Figure 4.4. We will discuss many more situations of negative work further in this unit.

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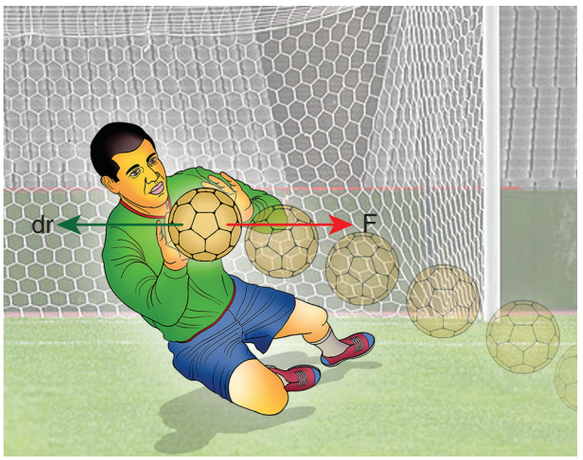
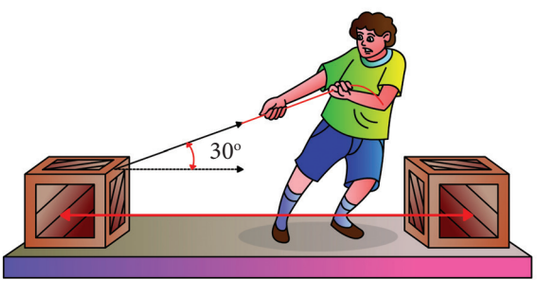


Figure 4.4 Negative work done

EXAMPLE 4.1

A box is pulled with a force of 25 N to produce a displacement of 15 m. If the angle between the force and displacement is 30°, find the work done by the force.



Solution

Force, F = 25 N
Displacement, dr = 15 m
Angle between F and dr, $\theta = 30^o$

Table 4.1 Angle (θ) and the nature of work

Angle (θ)	cosθ	Work
$\theta = 0^o$	1	Positive, Maximum
$0 < \theta < 90^o$ (acute)	$0 < \cos\theta < 1$	Positive
$\theta = 90^o$ (right angle)	0	Zero
$90^o < \theta < 180^o$	$-1 < \cos\theta < 0$	Negative
$\theta = 180^o$	-1	Negative, Maximum

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